OpenCaseLaw: A Verifiable Multilingual Citation Graph for Swiss Jurisprudence

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Abstract

We release **OpenCaseLaw**, an open, verifiable Swiss legal graph that joins court decisions, resolved case citations, statute references, legislative-history materials, scholarly commentaries, stable identifiers, and cryptographic provenance. The release contains **972,882** court decisions across **109 courts** in 28 jurisdictional layers (DE/FR/IT, 1875–2026), with **8.05 M** resolved citation tokens (92.9 % coverage of 8.66 M extracted tokens), expanded to **8.10 M** citation link edges (denominator taxonomy in Section 4). The graph is bridged to **5,516** federal and **15,722** cantonal laws, **362** CC-BY/CC-BY-SA commentaries, and **8,124** article-level Materialien anchors from federal statutes to Federal Council messages (Botschaften). Every decision receives a Swiss-native `cli:ch` identifier with an ECLI projection and inclusion-proof support against a 2026-05-21 RFC6962/OpenTimestamps integrity root. As descriptive characterization of the released graph, we report row-normalised cross-language citation flow: Italian-language Swiss decisions cite outside Italian 84.6 % of the time, French-language decisions 44.3 %, and German-language decisions 15.0 %; the aggregate cross-language share over the released resolved graph is 34.0 %. We report quality-control precision proxies (recorded-date chronology violations: 1.53 %, *not* a lower bound on precision), ship a 400-sample stratified audit set whose manual annotation is companion-paper scope, and explicitly delimit unresolved per-language resolution-rate confounders. Corpus CC0 on Hugging Face; code MIT. Snapshot 2026-05-21.

1 Introduction

Swiss legal NLP has lacked a public substrate joining the objects legal practitioners actually move between: court decisions, case citations, statute articles, legislative materials, doctrinal commentary, identifiers, and provenance. Existing Swiss datasets are valuable for tasks such as prediction, summarisation, translation, and metadata analysis, but they do not provide a cross-court graph that can be queried, audited, and cited as infrastructure. *OpenCaseLaw* fills that gap.

Given Switzerland's multilingual legal order, cross-language citation is expected: federal supreme-court decisions are authoritative in German, French, or Italian, and practitioners cite across those boundaries. The contribution is therefore the released graph that makes citation flow measurable, inspectable, and reusable at edge level across 109 courts and adjacent statute/Materialien layers. As one corpus characterization, the released resolved graph shows the following row-normalised language pattern:

Source language	Total resolved links	Same-lang	Cross-lang	Cross-lang share
German (de)	3,270,791	2,779,288	491,503	15.0 %
French (fr)	4,526,065	2,522,708	2,003,357	44.3 %
Italian (it)	305,380	47,106	258,274	**84.6 %**
Aggregate	8,102,236	5,349,102	2,753,134	34.0 %

We treat this matrix as descriptive corpus telemetry rather than a standalone result. It is still useful because it quantifies the resolved graph's language mix, exposes where per-language precision audit matters most, and gives downstream retrieval systems an edge-level map of the multilingual citation substrate. We do not claim a structural cause from this observation alone; per-source-language resolution-rate differentials remain a measurement-artefact concern (Section 8).

Why this substrate was missing. The existing open Swiss-law resources are either pre-training text [6], task benchmarks [5, 7, 10, 11], or court-specific metadata datasets such as the Swiss Federal Supreme Court Dataset (SCD) [1]. These resources are valuable, but they do not resolve a cross-court citation graph at scale, and they do not link case law to the Federal Council Botschaften that drive teleological statutory interpretation under Swiss civil-law doctrine. The infrastructure required for a reusable Swiss legal graph — federated cantonal-level decision coverage, multilingual docket normalisation, BGE/ATF/DTF prefix-form aliasing, pin-cite-aware resolution, article-level statute anchoring, and integrity proofs — did not exist as an open release.

Contributions. Four:

1. A resolved multilingual citation graph for Swiss jurisprudence at 92.9 % coverage, with a documented pin-cite-aware resolver, automated precision proxies, and a 400-sample stratified audit set queued for human adjudication.
2. A statute graph (11.28 M edges over 283,330 provisions) and an article-level Materialien bridge from federal statutory articles to Federal Council Botschaften.
3. A corpus characterization: row-normalised citation flow by source language (DE 15 %, FR 44 %, IT 85 % out-of-language), reported as descriptive telemetry for the released graph and subject to per-language resolution-rate confounders (Section 8).
4. A Swiss-native `cli:ch` identifier with a Council-of-EU ECLI projection in JSON-LD, plus an RFC-6962 Merkle root anchored on Bitcoin via OpenTimestamps and a per-decision inclusion-proof API.

The companion evaluation paper covers cross-lingual retrieval diagnostics with multi-annotator inter-annotator agreement, audit-rail adversarial probes, human calibration of the LLM grounding judge, and the manual annotation of the 400-sample citation-precision audit set.

2 Related work

Open legal corpora (international). The Caselaw Access Project [3] releases 6.7M US decisions; Harvard's Library Innovation Lab released a CAP citation graph as a static CSV in 2020.[1] CAP is single-jurisdiction (US) and monolingual; CAP's published graph is not paired with a statute graph, legislative-history bridge, or live retrieval API. The *Open Legal Data* platform [8] releases German court decisions with a citation network covering law and court nodes;

[1] Library Innovation Lab, *Caselaw Access Project Citation Graph*, 2020-04-22, `https://lil.law.harvard.edu/blog/2020/04/22/caselaw-access-project-citation-graph/`.

OLD is single-jurisdiction (Germany), monolingual, with no article-resolved bridge to legislative materials. *LeCNet* [2] introduces an Indian-judicial-context citation-network benchmark (26 k case-judgment nodes) for missing-citation recommendation. A very recent Ukrainian release extracts 502M citation links from 99.5M court-decision full texts, with topology and ontology analyses over a national monolingual corpus [9]. *Pile of Law* [4] and *MultiLegalPile* [6] are pretraining corpora and do not publish a resolved citation graph at this scale.

Swiss legal NLP benchmarks and datasets. Four task benchmarks on Swiss legal text precede our resource: *Swiss-Judgment-Prediction* [5] (binary outcome prediction); the *One Law, Many Languages* benchmark [11] (multi-task suite spanning document-to-document IR, court-view generation, leading-decision summarisation, citation extraction, classification across five languages); *Swiss Landmark Decisions Summarisation* [10] (20 k BGE landmark rulings paired with DE/FR/IT headnotes for cross-lingual summarisation, 1954–2024); and *SwiLTra-Bench* [7] (180 k aligned Swiss legal translation pairs across DE/FR/IT/RM/EN spanning federal laws, BGE headnotes, and Federal Supreme Court press releases). The Swiss Federal Supreme Court Dataset (SCD) records 120,368 Federal Supreme Court cases through 2023 with structured metadata, including citation and publication-status variables [1]. These resources target specific downstream tasks or one federal-court layer; they do not publish a resolved cross-court citation graph or a statute-anchored Materialien layer. The Federal Supreme Court layer represented in these resources is one of 28 jurisdictional layers we federate.

Comparison. Table 1 positions OpenCaseLaw against these prior releases. Inclusion criterion: open releases that publish a retrieval substrate over court-decision text — a citation graph, a statute-decision link layer, or a live retrieval endpoint — against which downstream RAG and legal-research pipelines can be built. Excluded: task benchmarks (SwiLTra, SLDS, OLML, SJP) and pretraining corpora (Pile of Law, MultiLegalPile), which publish task data or large text dumps rather than retrieval substrates; these are discussed in the prose above. SCD is a borderline case (it ships structured metadata over BGer but not a graph layer); we include it to make the criterion transparent.

Table 1: Comparison of selected open legal-data resources along dimensions relevant for retrieval-augmented Swiss legal research. "Citation graph" = resolved citation network (case-to-case unless noted); "Statute graph" = decision-to-statute-article edges with resolved targets; "Leg. history" = statute-article-to-legislative-history links (e.g. Botschaften, Hansard, committee reports) at article granularity; "Live API" = continuously served retrieval endpoint; "Cross-lang links" = the resource publishes a link layer where edges span ≥ 2 official languages (independent of whether the underlying corpus is multilingual). SCD covers DE/FR/IT Federal Supreme Court text but ships metadata variables, not a cross-language link layer. The table foregrounds retrieval-substrate features rather than task-benchmark coverage; pretraining corpora such as Pile of Law and MultiLegalPile, and Swiss task datasets such as SwiLTra/SLDS/OLML/SJP, are discussed in text.

Resource	Jurisdiction	Cross-lang links	Citation graph	Statute graph	Leg. history	Live API
Caselaw Access Project [3]	US	no	static CSV	no	no	no
Open Legal Data [8]	Germany	no	law/court nodes	no article graph	no	no
LeCNet [2]	India	no	26k bench	no	no	no
Ukrainian citation graph [9]	Ukraine	no	502M extracted links	no	no	no
SCD [1]	CH	no	metadata variables	no	no	no
OpenCaseLaw	CH (+ECHR Swiss)	yes	yes, 8.05M resolved	yes, 11.28M edges	yes, 8,124 anchors	yes

Table 2: OpenCaseLaw corpus overview. Snapshot 2026-05-21.

Quantity	Value
Total decisions	972,882
Distinct courts	109
Cantons + federal	28
Date range	1875–2026
German (de)	450,435 (46.3%)
French (fr)	441,673 (45.4%)
Italian (it)	80,774 (8.3%)

3 Corpus

The OpenCaseLaw corpus comprises **972,882 decisions** from **109 distinct courts** across **26 Swiss cantons plus federal and ECHR jurisdictions** (28 jurisdictional layers). Coverage spans **1875–2026** and three official languages: German (450,435 / 46.3 %), French (441,673 / 45.4 %), Italian (80,774 / 8.3 %). The federal Supreme Court (BGer) is refreshed hourly (Mon–Fri 05:00–16:00 UTC) via a dedicated Neuheiten poller; all other federations are refreshed by a nightly publish at 03:00 UTC. The federation aggregates source-specific scrapers and adapters covering federal courts (BGer, BGE, BVGer, BStGer, BPatGer), per-canton portals, the European Court of Human Rights (Swiss subset), historical archives, and federal regulatory authorities (FINMA, WEKO, and five others).

ECHR Swiss subset. The Swiss-relevant subset of European Court of Human Rights jurisprudence (**2,788 decisions**: 1,153 chamber + 93 grand-chamber + 230 committee + 835 HUDOC entries + 477 BGE-EGMR mirror) is integrated as a first-class federation. The subset matters because ECHR jurisprudence is directly applicable in Swiss litigation, and the official versions of ECHR-Switzerland judgments are typically French and English rather than German or Italian, which requires a multilingual retrieval substrate.

Completeness. Some source portals impose access restrictions (geographic, rate, or contractual). We document affected sources in the per-source coverage report, do not claim full coverage where access is restricted, and use only legally authorised collection methods. For closed historical archives where contemporaneous collection is no longer possible, we recover from publicly archived snapshots (e.g. Wayback Machine; ~903 BL ALS judgments).

Licensing. Court decisions in Switzerland are official published acts under URG/CopA Art. 5 (excluded from copyright). The corpus packaging and added metadata are released CC0 to the extent rights subsist. Scholarly commentaries retain their upstream terms (CC-BY-4.0 from OnlineKommentar.ch; CC-BY-SA-4.0 from OpenLegalCommentary.ch). The URG Art. 5 posture, the FADP (Swiss data protection) treatment of personal data, and the correction/removal-request procedure are documented in the public governance-and-removal policy at `https://github.com/jonashertner/caselaw-repo-1/blob/main/docs/governance-and-removal-policy.md`.

4 Multilingual citation graph

Denominator taxonomy. Counts in this section use one frozen 2026-05-21 snapshot. We use four distinct denominators:

Table 3: Citation graph. Edges extracted by full-text scanning every decision in the corpus; resolved means the cited target is identified in the corpus. Snapshot 2026-05-21.

Quantity	**Value**
Decisions analysed for outgoing citations	972,880
Citation edges (source $\rightarrow$ target reference)	8,663,860
Edges with target resolved to a corpus decision	8,047,620 (92.9%)
Distinct decisions cited at least once	261,677
Decisions cited $\geq$ 100 times	10,750
Decisions cited $\geq$ 1,000 times	975
Decisions cited $\geq$ 10,000 times	40

Table 4: Cross-lingual citation matrix. Rows are source decision languages; columns are cited (target) decision languages. Cells give resolved citation edges. Off-diagonal = 2,753,134 / 8,102,236 = 34.0% of resolved citations with identified source and target language.

	→ DE	**→ FR**	**→ IT**
DE →	2,779,288	459,563	31,940
FR →	1,939,860	2,522,708	63,497
IT →	168,741	89,533	47,106

- $E_{\text{extracted}}$ = **8,663,860**: distinct (source-decision, citation-token) pairs extracted by full-text scan over the corpus.
- E_{resolved} = **8,047,620**: subset of $E_{\text{extracted}}$ whose token resolves to $\geq$ 1 target decision (92.9 %).
- E_{link} = **8,102,236**: rows of the published `citation_targets` table, expanded for target multiplicity (a small fraction of citation tokens resolve to multiple targets, e.g. BGE form references that match several published BGEs by docket).
- $E_{\text{link,lang}}$ = **8,102,236**: subset of E_{link} where both source and target language are in {de, fr, it}; in this snapshot, all link rows satisfy this.

The cross-lingual share (34.0 %, Section 1) is computed over $E_{\text{link,lang}}$; the 92.9 % resolution rate is computed over $E_{\text{extracted}}$. The precision proxies below are computed over E_{link} = 8,102,236 on the same 2026-05-21 frozen graph, grouped by `match_type`; the cross-language matrix and the precision proxies share the same release denominator.

Resolution at scale. We extract citation tokens by full-text scanning every decision in the corpus (**972,880** decisions in the reference-graph snapshot; the FTS5 search index carries 2 more decisions ingested between graph rebuild and FTS5 swap), recovering $E_{\text{extracted}}$ = **8,663,860**. Resolution proceeds in three exact-match passes (general docket-normalisation; BGE/ATF/DTF prefix-form; bare BGE numeric form), augmented by a *pin-cite-aware fallback* for Swiss legal citations such as *BGE 125 V 352* that pinpoint into the body of a case starting at *BGE 125 V 351*. The pin-cite resolver matches each unresolved BGE-form reference to the nearest preceding first-page in the same volume + division within 30 pages, contributing **652,547** additional matches at a −0.10 confidence discount versus exact matches. The deployed 92.9 % resolution is a 17.7 pp lift over an internal pre-fix resolver baseline of 75.2 %, recovered by fixing two implementation bugs: docket-normalisation drift in 2007–2009 Bundesgericht dockets stored with a space (`8C 862_2008`) versus the underscore form (`8C_862_2008`) emitted by the citation extractor, and the pin-cite mismatch above.

Heavy-tailed centrality. The citation graph is heavy-tailed (Table 3): 40 decisions are cited 10,000+ times, 935 are cited 1,000–9,999 times, and most cited decisions sit in the long tail.

Table 5: Citation-resolution precision proxies computed over the 8,102,236 link-edge rows of the released graph (matching E_{link}, Section 4; re-run on the 2026-05-21 frozen snapshot). Date-sanity pass = % of pairs where target decision date is not after the source's; violations are recorded-date chronology inconsistencies and indicate likely false-positive resolutions (conditional on the recorded source/target dates themselves being correct; we preserve source-portal-provided dates, see Limitations), not a lower bound on precision. Self-cite = pairs where source = target. Conf. p50 = median resolver confidence per stratum. Full per-stratum manual precision adjudication is deferred to v2.0 (see `benchmarks/citation_precision_audit.py`).

match_type	n	date sanity	self-cite	conf p50
`docket_norm`	3,288,460	97.25%	0	0.99
`bge_bare`	2,944,527	99.75%	0	0.85
`bge_norm`	1,216,702	99.26%	0	0.75
`bge_pincite`	652,547	97.35%	0	0.75
Overall	8,102,236	98.47%	0	—

The single most-cited decision is BGE 125 V 351 (66,883 incoming citations after canonical-form aggregation across BGE id variants). Of the 50 most-cited decisions, 38 are DE-originals, 12 are FR, and none are IT — a distinct cut of the graph (citation reception, not direction).

Precision proxies. The 92.9 % resolution number is a *coverage* metric and does not bound precision. To characterise precision risk we compute three automated proxies (Table 5). The proxies surface *recorded-date chronology violations*: each detected case is a logical inconsistency *conditional on the source and target decision dates being themselves correct.* We preserve source-portal-provided dates (Section 8), so a violation is a strong but not infallible indicator of a false-positive resolution. Date-consistent, non-self pairs may still be false positives for reasons our automated proxies do not see (e.g., a docket-form citation that resolves to the wrong case with the same docket pattern). *Date-order sanity* (target decision not dated after source) finds violations in 1.53 % of E_{link} rows with both dates known (weighted across match types). This bounds precision on the released graph at *at most* 98.47 % on date-covered pairs conditional on the recorded dates; it does *not* establish a lower bound on precision.

The pin-cite stratum scores lowest (2.65 % violation rate), consistent with its heuristic nature; the exact-match strata (`bge_bare`, `bge_norm`) sit above 99.2 % pass. *Self-citation* is zero across all 8,102,236 rows. These proxies are necessary but not sufficient for a precision claim; we shipped a 400-sample stratified audit set (`benchmarks/citation_precision_sample_400.jsonl`) and an interactive auditor harness (`benchmarks/citation_precision_audit.py`) for human adjudication. The manual annotation pass is companion-paper scope.

Cross-language citation flow. Table 4 and Figure 1 report the language-by-language citation matrix over $E_{\text{link,lang}}$ = 8,102,236 as descriptive characterization of the released resolved graph. The aggregate cross-language share is **34.0 %**; row-normalisation by source language (Section 1, lead table) makes the per-language structure of the released graph visible: German-language decisions cite outside German in only **15.0 %** of their resolved links; French-language decisions cross language in **44.3 %**; Italian-language decisions cross language in **84.6 %**. FR→DE links (**1.94 M**) outweigh DE→FR (**0.46 M**) by **4.2**×; IT→DE/FR (**0.26 M** combined) outweighs IT→IT (**47 k**) by **5.5**×. These ratios are computed over the population of resolved link edges with both languages known; there is no sampling uncertainty, but there is measurement uncertainty from unresolved tokens and possible language-pair resolution bias. We do not claim a structural cause from this observation alone — unresolved citation tokens have no target language, and the resolver was tuned on BGE-form citations, which are predominantly DE-target, so a per-source-language or per-language-pair resolution-rate differential could partly

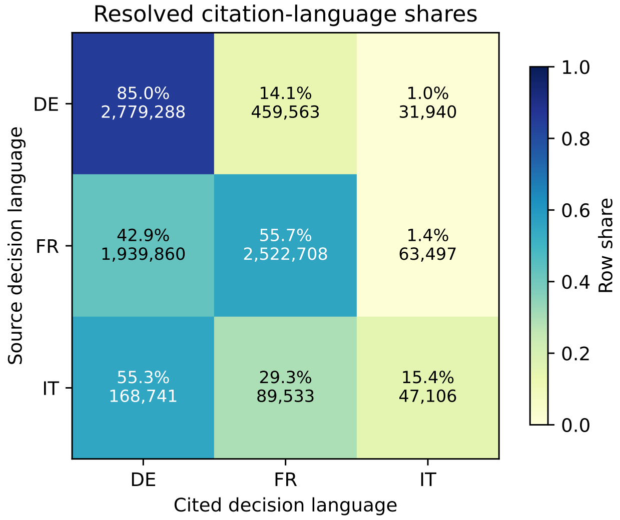

Figure 1: Row-normalised citation-language flow over the released resolved graph. Each cell gives the share within a source-language row and the underlying resolved link-edge count.

inflate the FR and IT cross-lingual shares (Section 8); the proper control is a per-language-pair precision audit (queued, Section 8).

Table 6: Top 15 most-cited decisions in the corpus. In-degree counts citing decisions (with multi-citing pairs counted once per source). Snapshot 2026-05-21.

Decision ID	**Reporter form**	**In-degree**
`bge_125 V 351`	BGE 125 V 351	66,883
`bge_122 V 157`	BGE 122 V 157	40,452
`bge_134 V 231`	BGE 134 V 231	36,622
`bge_130 V 343`	BGE 130 V 343	26,917
`bge_125 V 256`	BGE 125 V 256	24,168
`bge_125 V 193`	BGE 125 V 193	22,577
`bge_126 V 353`	BGE 126 V 353	22,415
`bge_115 V 133`	BGE 115 V 133	20,904
`bge_135 V 465`	BGE 135 V 465	20,025
`bge_137 V 210`	BGE 137 V 210	19,829
`bge_126 V 75`	BGE 126 V 75	19,713
`bge_138 III 374`	BGE 138 III 374	18,314
`bge_130 III 321`	BGE 130 III 321	18,191
`bge_125 V 413`	BGE 125 V 413	17,566
`bge_133 II 249`	BGE 133 II 249	16,470

5 Statute graph and Materialien bridge

Statute graph. We resolve **11,276,187 decision-to-statute edges** touching **283,330 distinct provisions**. The statute layer is grounded in two parallel mirrors: a federal layer of **5,516 SR-numbered Fedlex laws** (400,405 article rows summed across the parallel DE/FR/IT versions; ∼133 k articles per language) and a cantonal layer of **15,722 laws** (353,437 articles) covering all 26 cantons via direct portal scraping (LexWork, 18 cantons; SIL, 2 cantons; ZH OpenData, ZH; TI RL, TI), with LexFind PDF fallback supplementing 4 cantons for laws missing from their primary portals. Top-referenced provisions (Table 8, after alias canonicalisation across 20 major DE/FR/IT statute-abbreviation pairs) are dominated by federal-court procedural code: *BGG/LTF* Art. 100 (**205,725** decisions, 30-day appeal deadline), Art. 42 (**201,897**, statement-of-grounds requirement), and Art. 113 (**122,514**, subsidiary constitutional complaint). Beyond procedural code, *ATSG/LPGA* Art. 61 (social-insurance procedure, 85,207) and *BV/Cst./Cost.* Art. 29 (procedural fundamental rights, 79,115) lead the non-BGG/LTF substantive provisions. Alias canonicalisation is lexical only; temporal validity — mapping a 1998 *OG* reference to its current SR location, or distinguishing repealed-vs-current article versions — is left for future work.

Table 7: Statute and doctrinal graph layers. Decision-to-statute edges plus federal/cantonal mirrors plus Materialien (Federal Council Botschaften) and open commentaries.

Quantity	**Value**
Decision → statute edges	11,276,187
Distinct provisions referenced	283,330
Federal laws (Fedlex SR series)	5,516
Federal articles (DE/FR/IT parallel)	400,405
Cantonal laws (all 26 cantons; 22 direct, 4 LexFind fallback)	15,722
Cantonal articles	353,437
Materialien (Botschaft documents, DE/FR/IT)	5,292
Materialien paragraphs (full-text)	381,711
Article-anchored Materialien links	8,124
Open commentaries (CC-BY/CC-BY-SA)	362

Table 8: Top 15 most-referenced statutory provisions. Alias-canonicalised across DE/FR/IT abbreviations of 20 major Swiss federal statutes; decisions citing multiple language forms of the same provision in one document are counted once.

Law	**Article**	**Decisions citing**
BGG/LTF	Art. 100	205,725
BGG/LTF	Art. 42	201,897
BGG/LTF	Art. 113	122,514
BGG/LTF	Art. 66	97,701
ATSG/LPGA	Art. 61	85,207
BV/Cst./Cost.	Art. 29	79,115
VwVG/PA	Art. 63	70,498
BGG/LTF	Art. 106	65,798
BGG/LTF	Art. 72	64,793
VwVG/PA	Art. 52	64,246
BGG/LTF	Art. 82	62,398
BGG/LTF	Art. 74	62,217
StPO/CPP	Art. 428	59,567
BGG/LTF	Art. 95	58,055
ZPO/CPC	Art. 106	56,092

Materialien graph: candidate discovery. A distinguishing feature of Swiss civil-law interpretation is the *teleological* reading of statutes via *Materialien* — legislative-history documents (Federal Council messages, parliamentary debates) that record legislative intent. We discover Materialien via the Fedlex JOLUX SPARQL endpoint with filter `typeDocument` = 23 (Federal Council Botschaft). The endpoint returned candidate documents from 2003 (the earliest year for which Fedlex JOLUX exposes a parliamentary-chain anchor) to the 2026-05-21 snapshot date. We ingest **5,292** Botschaft documents at paragraph granularity (**381,711 paragraphs total**) from Akoma Ntoso XML where available, with PDF fallback for older issues. Per-language coverage reflects the upstream Botschaft's own publication: not all Botschaften are translated into all three official languages by the Federal Council.

Materialien graph: anchoring. We anchor Botschaft documents to specific `(SR, article)` pairs by traversing the official parliament chain: `parliamentDraftId` → enacted `oc` (Official Compilation) → codified `cc` (consolidated codification) → `SR` (Systematic Register). This yields **8,124** article-to-Botschaft anchors (DE 2,798, FR 2,742, IT 2,584). The link is at the Botschaft-document→SR-article level (not paragraph→article); finer-grained paragraph-to-article anchoring is left for future work. Known failure modes: (i) Botschaften that propose amendments which were rejected by Parliament have no enacted-oc side and therefore no SR anchor; (ii) Botschaften that bundle amendments to multiple statutes generate one link per affected statute, which inflates the link count relative to the document count; (iii) article-renumbering events in the codification step (rare but observed) currently anchor the Materialien link to the post-renumbering article identifier only. We have not yet computed an end-to-end precision audit for the article-to-Botschaft anchoring, so the present release treats this layer as a high-recall legislative-history discovery index rather than as an adjudicated alignment benchmark. The layer is queryable via search_materialien / get_materialien retrieval interfaces and surfaces verbatim legislative-history text under anchored statute articles.

Commentaries. We bridge **362 scholarly commentaries** (CC-BY-4.0 from OnlineKommentar.ch; CC-BY-SA-4.0 from OpenLegalCommentary.ch) into the graph, anchored by SR number plus article. The doctrinal layer is currently shallow but growing under the upstream commentaries' own publication cadence.

6 Identifier and provenance layer

Layered identifier (cli:ch + ECLI). Every decision carries a Swiss-native canonical identifier cli:ch with a Council-of-EU ECLI projection, both surfaced in the per-decision Schema.org/LegalCase JSON-LD. cli:ch makes five features of Swiss jurisprudence first-class that an ECLI-only scheme flattens: (i) *trilingualism* — DE/FR/IT versions of a BGE are legally equally authentic texts (Art. 14 PublG), and cli:ch treats language as a first-class axis via ?lang=; (ii) *cantonal sovereignty* — 26 distinct judicial systems remain visible (cli:ch:zh:obergericht:...) rather than collapsing into an opaque three-letter court code; (iii) *pinpoint addressing* — Erwägung-level citation as a first-class fragment (#e-2.3), matching the Schweizer-Citation convention; (iv) *chamber granularity* — Obergericht, Verwaltungsgericht, etc. remain explicitly named; (v) *versioning* — @YYYY-MM-DD addresses a specific anonymisation version when needed. The compact form is

```
cli:ch:<court>[:<chamber>]:<docket>[@<version>][?lang=<l>][#<pinpoint>].
```

The mapping to ECLI is a pure function (cli_ch_to_ecli); the projection is many-to-one (lossy on language, pinpoint, version, chamber detail). An HTTP resolver at https://opencaselaw.ch/cli/ch/ plus the court/chamber/docket path redirects to the canonical decision page; browsers preserve fragment pinpoints across the 302.

Cryptographic provenance (RFC-6962 + OpenTimestamps). Every daily publish computes an RFC-6962 Merkle tree over all decisions in the corpus. Each leaf commits to the tuple (decision_id, cli:ch, ECLI, content_hash, decision_date). The root (32 bytes, SHA-256-based, hash-domain-separated per the Certificate-Transparency RFC) is committed to the public Git repository at docs/integrity/<YYYY-MM-DD>.root and anchored on the Bitcoin blockchain via OpenTimestamps (.root.ots, $\sim$945 bytes). The first root, over 972,882 decisions on 2026-05-21, is 1b597b92...81f599. A GET /api/integrity/<decision_id> endpoint returns an inclusion proof in $O(\log n) \approx 20$ sibling hashes, computed in milliseconds against a memoised subtree cache. Any client can independently verify that a given decision was in the corpus on a given date without trusting the server: leaf $\rightarrow$ inclusion-proof walk $\rightarrow$ root $\rightarrow$ OpenTimestamps proof $\rightarrow$ Bitcoin block. This makes per-decision inclusion verification a public API surface rather than an internal release-engineering artifact.

Standards proposal. The identifier convention, the JSON-LD metadata schema, Akoma Ntoso for legislative materials, Markdown-with-pinpoint-anchors for full text, the resolved-citation conventions used here, and the Merkle-plus-OpenTimestamps provenance layer are consolidated as a six-bullet open-standards proposal at the *Standards* URL in Section 10. Each bullet is independently adoptable.

7 Dataset health

The release is protected by a four-layer quality framework: L1 function tests (624 pytest cases across 60 modules, every CI push); L2 build-time pipeline guards (codified `_normalize_*` passes in `build_fts5.py`, every nightly rebuild); L3 dataset audit (52 dataset-level `check_*` functions across 18 axes; per-check JSON published at `https://opencaselaw.ch/quality.json`, visualisation at `https://opencaselaw.ch/quality.html`, both refreshed every nightly rebuild); L4 production smoke (every 5 min, 6 endpoint probes, $p_{95} < 2$ s). A critical regression detected by L3 *blocks the nightly publish* before the git-push step; critical conditions include a total-count cliff >1 %, schema break, court not in registry, future-date count >50, and cross-DB row-count mismatch >0.5 %. Production incidents detected by L3 are codified as permanent regression tests in the source repository.

8 Limitations

Citation precision. The 92.9 % *resolution* number is a coverage metric, not a precision bound. The three automated proxies (date sanity 98.47 %, self-citation 0 %, confidence-percentile distributions per `match_type`) give a recorded-date-chronology-violation floor and an upper bound on date-sanity-visible precision *conditional on the recorded decision dates being correct*; they do *not* give a lower bound on precision. They do not substitute for human adjudication of citation correctness. A 400-sample stratified audit set is shipped (`benchmarks/citation_precision_audit.py`, `benchmarks/citation_precision_sample_400.jsonl`); we have not yet annotated it.

Temporal validity of statute references. The statute graph maps decision references to the *current* statute text. A 1998 decision referencing *OG* (the pre-2007 Federal Justice Act, since superseded by BGG/LTF) currently resolves to the current SR location of the surviving content but does not distinguish the original-vs-superseded version. Versioned statute resolution is future work.

Source pseudonymisation is inherited. Most federal decisions are pseudonymised by the issuing court (parties appear as *A.*, *X. AG*, etc.); some cantonal decisions are published in unredacted form by the issuing court itself. We neither re-pseudonymise material the court chose to publish nor de-pseudonymise material the court chose to redact. Removal requests are honoured per the documented governance policy.

Resolution rate. The 92.9 % resolution rate represents a 17.7 pp lift over the prior release's 75.2 % but a 0.6 pp regression from the previous snapshot's 93.5 %; the regression is real and reflects citations to recently-added decisions whose targets are not yet in the corpus. Future v2.0 resolver work will narrow this.

Per-language resolution-rate confounder for the citation-flow matrix. The 34 % cross-lingual share is computed over the released resolved graph (8.10 M edges with both source and target languages known). If resolution rates differ systematically by language pair — for example, if FR-source decisions citing DE-targets resolve at higher rates than FR-source decisions citing FR-targets — the share could be biased relative to the (unobserved) ground-truth citation distribution. Our pin-cite-aware resolver was designed and tuned on BGE-form citations (which are disproportionately DE-original), so a residual bias toward higher DE-target resolution is plausible. The 400-sample stratified precision audit (above) is the route to a language-pair-controlled correction. Conditional on such a bias being present, our hypothesised direction is attenuation of the FR-to-FR and IT-to-IT diagonals and amplification of cross-DE flows; in that case the 34 % aggregate could be an upper bound rather than an underestimate. Both the magnitude and the direction await the language-pair precision audit, which the companion paper will report.

Snapshot-date semantics. Source portals sometimes expose scheduled or near-future publication dates. We preserve source-provided decision dates rather than rewriting them to the snapshot date. The dataset-health gate therefore treats large future-date spikes as critical scraper regressions but does not drop otherwise plausible near-future rows automatically.

Scope of this paper. We separate *resource release* (this paper) from *evaluation diagnostics* (companion paper). The cross-lingual citation-flow matrix reported here characterizes the released resolved graph; it is not the resource contribution itself. The cross-lingual retrieval diagnostic, the audit-rail framework, and the prior-only-vs-RAG-augmented bench are evaluation contributions that depend on this resource and will be reported once their multi-annotator and human-judge-calibration components are complete.

9 Discussion

What the resource enables. With this release, researchers can study cross-language citation flow as a first-class object, train retrievers and generators on a substrate where every resolved citation has a target identifier and every statute reference is anchored to an article, and build legal-RAG systems whose outputs are verifiable against a Bitcoin-anchored corpus snapshot. The Materialien bridge in particular opens teleological-interpretation research by exposing programmatic article-level access to Botschaft documents, subject to the anchoring limitations above.

Generalisation. The architecture (cross-court federation, pin-cite-aware resolver, cli:ch + ECLI identifier layer, RFC-6962 + OpenTimestamps provenance) is jurisdiction-agnostic. Adaptation to other multilingual civil-law systems (e.g. Belgium, Quebec, South Africa) would require local citation grammars, statute-version semantics, source-access policy, and legislative-history normalisation; the Swiss specifics in this release are localisation, not architecture.

Open questions. Three lines of follow-up work motivated by the resource: (i) precision-controlled measurement of the cross-lingual citation share with a multi-annotator stratified audit; (ii) versioned statute resolution so a 1998 *OG* reference resolves to the contemporaneous statute text rather than the current SR location; (iii) cross-lingual retrieval evaluation with lawyer-authored held-out queries (companion paper).

10 Open access and reproducibility

All artefacts are publicly available:

- Corpus: Hugging Face `https://huggingface.co/datasets/voilaj/swiss-caselaw` (CC0)
- Code: `https://github.com/jonashertner/caselaw-repo-1` (MIT)
- Live MCP/REST: `https://mcp.opencaselaw.ch` (SSE; OpenAPI at `/api/openapi.copilot.json`)
- Word add-in (lawyer-facing client): `https://opencaselaw.ch/word`
- Dashboards: `https://opencaselaw.ch`, `https://opencaselaw.ch/quality.html`
- Standards proposal: `https://opencaselaw.ch/standards/`
- Integrity / Merkle root and OpenTimestamps anchor: `https://opencaselaw.ch/integrity/`

Reproducibility capsule. The numerical claims in this paper are generated from the frozen table snapshot at `docs/paper/p1-resource/tables/corpus_graph_stats.json`; tables are produced by `docs/paper/v3/scripts/build_tables.py`, and Figure 1 by `docs/paper/p1-resource/scripts/build_figures.py`. Citation-precision proxies are stored at `benchmarks/citation_precision_proxies.json`; the unannotated human-audit sample is `benchmarks/citation_precision_sample_400.jsonl`. The 2026-05-21 integrity manifest is `docs/integrity/2026-05-21.json`; the full RFC6962-SHA256 root is stored in `docs/integrity/2026-05-21.root` and begins `1b597b92`. The archival DOI and submission git tag will be minted at submission, after this manuscript and the release artifacts are frozen.

Companion paper (evaluation diagnostics, in preparation) covers the cross-lingual retrieval diagnostic with multi-annotator inter-annotator agreement, the audit-rail adversarial probe results, and human calibration of the LLM judge used in the closing-audit grounding rail.

References

[1] Florian Geering and Jakob Merane. Swiss federal supreme court dataset (SCD), 2024. Version 2023-5.

[2] Pooja Harde, Bhavya Jain, and Sarika Jain. Lecnet: A legal citation network benchmark dataset. In *Proceedings of the 1st Workshop on NLP for Empowering Justice (JUST-NLP 2025)*, pages 18–28. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.justnlp-main.4.

[3] Harvard Law School Library. Caselaw access project. `https://case.law/`, 2018.

[4] Peter Henderson, Mark Krass, Lucia Zheng, Neel Guha, Christopher D. Manning, Daniel Jurafsky, and Daniel E. Ho. Pile of law: Learning responsible data filtering from the law and a 256gb open-source legal dataset. *Advances in Neural Information Processing Systems 35, Datasets and Benchmarks Track*, 2022.

[5] Joel Niklaus, Ilias Chalkidis, and Matthias Stürmer. Swiss-judgment-prediction: A multilingual legal judgment prediction benchmark. In *Proceedings of the Natural Legal Language Processing Workshop 2021*, pages 19–35. Association for Computational Linguistics, 2021. doi: 10.18653/v1/2021.nllp-1.3.

[6] Joel Niklaus, Veton Matoshi, Matthias Stürmer, Ilias Chalkidis, and Daniel E. Ho. Multilegalpile: A 689gb multilingual legal corpus. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15077–15094. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.805.

[7] Joel Niklaus, Jakob Merane, Luka Nenadic, Sina Ahmadi, Yingqiang Gao, Cyrill A. H. Chevalley, Claude Humbel, Christophe Gösken, Lorenzo Tanzi, Thomas Lüthi, Stefan Palombo, Spencer Poff, Boling Yang, Nan Wu, Matthew Guillod, Robin Mamic, Daniel Brunner, Julio Pereyra, and Niko Grupen. SwiLTra-Bench: The Swiss legal translation benchmark. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14894–14916. Association for Computational Linguistics, 2025. URL `https://aclanthology.org/2025.acl-long.725/`. 180k aligned Swiss legal translation pairs across DE/FR/IT/RM/EN spanning laws, BGE headnotes, and Federal Supreme Court press releases; introduces SwiLTra-Judge for LLM-based eval aligned with human experts.

[8] Malte Ostendorff, Till Blume, and Saskia Ostendorff. Towards an open platform for legal information. In *Proceedings of the ACM/IEEE Joint Conference on Digital Libraries (JCDL)*, 2020. Open Legal Data platform; arXiv:2005.13342.

[9] Volodymyr Ovcharov. Automatic construction of a legal citation graph from 100 million ukrainian court decisions: Large-scale extraction, topological analysis, and ontology-driven clustering. *arXiv preprint arXiv:2605.15362*, 2026. doi: 10.48550/arXiv.2605.15362.

[10] Luca Rolshoven, Vishvaksenan Rasiah, Srinanda Brügger Bose, Sarah Hostettler, Lara Burkhalter, Matthias Stürmer, and Joel Niklaus. Unlocking legal knowledge: A multilingual dataset for judicial summarization in switzerland. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 15382–15411. Association for Computational Linguistics, 2025. URL `https://aclanthology.org/2025.findings-emnlp.832/`. SLDS dataset: 20,000 Swiss Federal Supreme Court rulings with DE/FR/IT headnotes, CC-BY-4.0, https://huggingface.co/datasets/ipst/slds.

[11] Ronja Stern, Vishvaksenan Rasiah, Veton Matoshi, Srinanda Brügger Bose, Matthias Stürmer, Ilias Chalkidis, Daniel E. Ho, and Joel Niklaus. One law, many languages: Benchmarking multilingual legal reasoning for judicial support. *arXiv preprint arXiv:2306.09237*, 2024.